

Awakening the Enemy or Silencing Them Forever: Innovative Approaches to Combat Dormant Cancer Cells and Prevent Metastatic Relapse

Metastatic relapse occurring years after treatment of the primary tumor accounts for the majority of cancer-related deaths (Dillekås et al., 2019). This late recurrence is largely driven by the reactivation of dormant cancer cells (DCCs), a subpopulation of cancer cells that enter a temporary G0-G1 cell cycle-arrested state, also known as a reversible non-proliferative (quiescent) state (Hadfield, 1954). The dissemination of DCCs to distant sites can occur not only during early stages of cancer progression but also after the administration of conventional treatments as their non-proliferative state allows them to resist treatments that primarily target rapidly dividing cancer cells. Therefore, targeting DCCs post-primary therapy is critical to prevent relapse and improve prognosis in patients. This opinion piece will explore potential strategies for addressing dormant cancer cells and propose the most promising approach.

The ‘sleeping strategy’ focuses on maintaining DCCs in a dormant state, preventing reactivation, and, if possible, eradicating them while they remain quiescent. Dormancy can be maintained by either repressing proliferative signaling or upregulating dormancy pathways. Studies in basal cell carcinoma (BCC) have shown that initial treatment with the Hedgehog signaling inhibitor vismodegib often results in relapse due to a small, slow-cycling LGR5⁺ subpopulation with active Wnt signaling. In this context, Wnt inhibition can effectively silence this residual population and prevent relapse (Sánchez-Danés et al., 2018). However, such strategies are tumor-type-specific, and a more pan-cancer approach to target dormant cells is needed. CDK4/6 inhibitors, which prevent Rb phosphorylation and induce G1 cell-cycle arrest, have been shown to enforce cytostasis and could therefore be used to prevent cell cycle re-entry of DCCs (O’Leary et al., 2016). While CDK4/6 inhibitors are currently part of the standard of care for advanced breast cancer, their potential extends to other malignancies. Notably, sequential administration of CDK4/6 inhibitors following DNA-damaging chemotherapy was shown to improve the recovery of pancreatic ductal adenocarcinoma by enforcing G1 arrest and repressing homologous recombination repair, thereby preventing recovery from genotoxic stress. This effect may be potentially correlated with the maintenance of a dormant-like state in residual cancer cells (Salvador-Barbero et al., 2020). Moreover, a high p38/ERK ratio is a dormancy marker, and therefore activating p38 or downregulating ERK can result in the permanent arrest of dormant cells (Santos-de-Frutos and Djouder, 2021). Beyond maintaining dormancy, targeting survival mechanisms offers a strategy to eradicate dormant cells. Studies have found that residual pancreatic cancer cells surviving KRAS or c-MYC ablation rely on IGF1/AKT signaling for survival and its inhibition reduces recurrence (Rajbhandari et al., 2017). Similarly, in non-small cell lung cancer, quiescent cancer stem cells depend on Bcl-XL and treatment with the Bcl-XL inhibitor ABT-737 eliminates them effectively (Zeuner et al., 2014).

On the opposite side, the ‘awakening strategy’ aims to reactivate DCCs, forcing them to re-enter the cell cycle and become resensitized to anti-proliferative therapies. This approach relies on identifying and triggering intrinsic cellular pathways that mediate dormancy exit. In a mouse model of acute myeloid leukemia (AML), granulocyte colony-stimulating factor (G-CSF) was shown to induce cell cycle re-entry of quiescent AML stem cells, thereby enhancing their elimination when combined with chemotherapy (Saito et al., 2010). Similarly, interferon- α (IFN α) treatment has been found to activate dormant hematopoietic stem cells, increasing their susceptibility to 5-fluorouracil treatment (Essers et al., 2009).

From my own perspective, the ‘sleeping strategy’ is favored as the ‘awakening strategy’ involves rewiring of the dormant cells to a proliferative state which could increase the metastatic burden and if not well-treated, may result in an earlier and more aggressive relapse. Furthermore, I propose that combining ‘sleeping strategy’ with novel therapies, such as immunotherapies, represents a promising approach for treating DCCs. Studies suggest that immune evasion by DCCs may be due to limited interactions between DCCs and antigen-specific T cells, a challenge that could be addressed through T-cell immunotherapies (Goddard et al., 2024). With the development of artificial intelligence, AI could help predict where the DCCs are most likely to relocate for specific tumor types, enabling chances for proactive prevention. With the advancement of precision oncology, a personalized approach tailored to each patient’s specific molecular cancer type could be integrated: a ‘magical bullet’ containing modified T cells targeting neoantigens identified from the primary tumor could be delivered directly to the predicted sites of potential disseminated DCC niches and together with ‘sleeping strategy’ to prevent metastatic relapse.

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